Omoze Idehenre
PKI Week 4 Labs
3/31/2026

Lab 1 lab-01-cert-output.png

```
        Serial Number:
            57:a3:7e:67:bd:27:4c:80:09:b0:ef:35:88:62:5b:3d
    Signature Algorithm: ecdsa-with-SHA256
        Issuer: C=US, O=Google Trust Services, CN=WE2
        Validity
            Not Before: Mar 16 08:36:41 2026 GMT
            Not After : Jun  8 08:36:40 2026 GMT
        Subject: CN=*.google.com
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:7e:fe:95:89:63:5a:8b:e9:54:4f:16:0c:e4:a0:
                    0f:06:b9:e4:f6:b9:03:50:85:9c:8c:d5:d6:9a:5f:
                    79:d9:57:35:b4:79:80:ec:94:a9:15:4f:5e:f9:92:
                    e0:81:9b:a7:fe:66:9d:50:c3:27:91:2f:36:e7:3e:
                    8e:78:f2:0c:29
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        X509v3 extensions:
            X509v3 Key Usage: critical
                Digital Signature
```

Lab 2 lab-02-trust-store.png

```
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify /etc/ssl/cert.pem
/etc/ssl/cert.pem: OK
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```

```
Omozes-MacBook-Pro:convert-formats omozeidehenre$ openssl x509 \
> -in labs/week-04/submissions/trust-store-inspection/root_cas.pem \
> -noout -subject -issuer -enddate
subject= /C=US/ST=Arizona/L=Scottsdale/O=GoDaddy.com, Inc./CN=Go Daddy Root Certificate Authority - G2
issuer= /C=US/ST=Arizona/L=Scottsdale/O=GoDaddy.com, Inc./CN=Go Daddy Root Certificate Authority - G2
notAfter=Dec 31 23:59:59 2037 GMT
Omozes-MacBook-Pro:convert-formats omozeidehenre$ subject= /CN=DigiCert Global Root CA/OU=www.digicert.com/O=DigiCert Inc/C=US
-bash: /CN=DigiCert: No such file or directory
Omozes-MacBook-Pro:convert-formats omozeidehenre$ issuer= /CN=DigiCert Global Root CA/OU=www.digicert.com/O=DigiCert Inc/C=US
-bash: /CN=DigiCert: No such file or directory
Omozes-MacBook-Pro:convert-formats omozeidehenre$ notAfter=Nov 10 00:00:00 2031 GMT
-bash: 10: command not found
Omozes-MacBook-Pro:convert-formats omozeidehenre$
```

Lab 3 lab-03-verification.png

```
>   -CAfile labs/week-04/submissions/install-validate/test-root-ca.crt \
>   labs/week-04/submissions/install-validate/test-signed.crt
labs/week-04/submissions/install-validate/test-signed.crt: OK
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify \
>   labs/week-04/submissions/install-validate/test-signed.crt
CN = test.cvi-lab.local, O = CyberVisionaries Institute, C = US
error 20 at 0 depth lookup:unable to get local issuer certificate
labs/week-04/submissions/install-validate/test-signed.crt: verification failed: 20 (unable to get local issuer certificate)
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify \
>   -CAfile labs/week-04/submissions/install-validate/test-root-ca.crt \
>   labs/week-04/submissions/install-validate/test-signed.crt
labs/week-04/submissions/install-validate/test-signed.crt: OK
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify \
>   labs/week-04/submissions/install-validate/test-signed.crt
CN = test.cvi-lab.local, O = CyberVisionaries Institute, C = US
error 20 at 0 depth lookup:unable to get local issuer certificate
labs/week-04/submissions/install-validate/test-signed.crt: verification failed: 20 (unable to get local issuer certificate)
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```

```
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify \
>   -CAfile labs/week-04/submissions/install-validate/test-root-ca.crt \
>   labs/week-04/submissions/install-validate/test-signed.crt
labs/week-04/submissions/install-validate/test-signed.crt: OK
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify \
>   labs/week-04/submissions/install-validate/test-signed.crt
CN = test.cvi-lab.local, O = CyberVisionaries Institute, C = US
error 20 at 0 depth lookup:unable to get local issuer certificate
labs/week-04/submissions/install-validate/test-signed.crt: verification failed:
20 (unable to get local issuer certificate)
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify \
>   -CAfile labs/week-04/submissions/install-validate/test-root-ca.crt \
>   labs/week-04/submissions/install-validate/test-signed.crt
labs/week-04/submissions/install-validate/test-signed.crt: OK
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify \
>   labs/week-04/submissions/install-validate/test-signed.crt
CN = test.cvi-lab.local, O = CyberVisionaries Institute, C = US
error 20 at 0 depth lookup:unable to get local issuer certificate
labs/week-04/submissions/install-validate/test-signed.crt: verification failed:
20 (unable to get local issuer certificate)
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```